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Boltzmann Temperature Schedules, Reward Scaling, and Q-Learning Convergence

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Introduction

Boltzmann or softmax exploration uses a temperature parameter to trade off exploration and exploitation. High temperature makes action choice close to random, while low temperature makes the policy focus on the highest-value action, so practical use depends on how temperature is annealed over time and how Q-values are scaled. (11 sources)

Boltzmann exploration defines the policy by turning action values into a softmax distribution, so actions with larger estimated Q-values get larger selection probability, but lower-value actions can still be chosen. The key control is the temperature parameter: when temperature is high, action probabilities become close to uniform and behavior is mostly exploratory; when temperature is low, the distribution concentrates on the best-valued action and behavior becomes closer to greedy exploitation. Because of this, many descriptions recommend starting with a high temperature early in learning and then reducing it gradually so the agent explores broadly at first and exploits more later. This basic setup is the background for the main issues in the rest of the answer: what kinds of annealing schedules are needed, what goes wrong if the temperature drops too fast or too slow, how reward or Q-value scaling changes the softmax probabilities, and how these exploration choices relate to standard Q-learning convergence conditions on learning rates and repeated state-action visitation. (Sharaf et al., 2017) (Nocera et al., 2014) (Rahmati et al., 2019) (Wei et al., 2019) (Li et al., 2020) (Nguyen et al., 2020) (Amin et al., 2021) (Xiao et al., 2025) (Zhang, 2026) (That et al., 2026) (Ameen et al., 2026)

Evidence

(Sharaf et al., 2017)

Structured Prediction via Learning to Search under Bandit Feedback

> *"High temperatures cause the actions to be all (nearly) equiprobable. Low temperatures cause a greater difference in selection probability for actions that differ in their value estimates."*

(Nocera et al., 2014)

The role of intrinsic motivations in attention allocation and shifting

> *"Different exploration policies can be introduced to select the action to be executed trying to balance exploration and exploitation. Analogously to Di (Nocera et al., 2012), in this paper we consider a Softmax method that selects the action to be executed through a Boltzmann distribution (Sutton and Barto, 1998):"*

> *"Here, the temperature τ controls the exploration strategy: the higher the temperature, the closer the strategy is to a random policy (exploration); the lower the temperature, the closer the strategy is to Q(s, a) maximization (exploitation)."*

(Rahmati et al., 2019)

UW-MARL: Multi-Agent Reinforcement Learning for Underwater Adaptive Sampling using Autonomous Vehicles

"In Boltzmann exploration, the probability of selecting an action, a, in a given state, s, is directly related to its Q-value as π t (s, a) = e Q t (s,a)/T / a ' Æ Q t (s,a ')/T . Here, the parameter (temperature T) decides the amount of exploration or exploitation; if T is large, all actions have almost equal probability resulting in pure exploration whereas when T tends to 0 the optimal action has the highest probability, resulting in pure exploitation. Hence, it is desired to have a high value for T in the beginning and reduce it gradually to get optimal performance (similar variation is needed for learning rate, α)."

(Wei et al., 2019)

Multiobjective model-free learning for robot pathfinding with environmental disturbances

"At the beginning of the learning process, the temperature is high, so all actions will be selected with almost equal probability regardless of their relative values. To address the issue of 'Dimension 1: Initially explore more, gradually exploit more', the temperature should be annealed down over time during the learning process. As a result, the learning agent will most likely choose higher value actions, but it is not guaranteed. Thus, the temperature parameter can decide to what extent the softmax distribution will be spread. At the end of the learning process, the parameter will approach zero, so all actions are sparsely distributed, and the action selection converges towards a greedy method."

(Li et al., 2020)

Dynamic Colocation Policies with Reinforcement Learning

"Softmax exploration uses the Boltzmann distribution function to determine the probability that an agent selects action a in state s. T is the temperature factor in the Boltzmann distribution. As T 0, the agent explores less."

(Nguyen et al., 2020)

Q-Learning with Differential Entropy of Q-Tables

"where, T is the temperature affecting the randomness of the actions. The higher the value of T , the more exploration behaviour the agent performs. The temperature is reduced gradually during experiments to facilitate more greedy actions."

(Amin et al., 2021)

A Survey of Exploration Methods in Reinforcement Learning

"One of the most important exploration methods in this category is the Softmax action selection method (Bridle, 1989). In this method, the greedy action at the current state is selected with the highest probability, while the other actions are given the probability of being selected according to their estimated values. The most commonly used formalism for performing Softmax action selection is the Boltzmann distribution. The early use of Boltzmann distribution in exploration was by Watkins (1989); (Lin, 1992) and Barto et al. (1991). In the Boltzmann exploration approach"

"where T m > 0 is called the temperature and controls how frequent the agent will choose random actions as opposed to the best actions a t = arg max a Q t (s, a). If T m decreases, the probability of generating the action with the highest expected reward a t increases, leading to a decrease in the probability of selecting other actions and hence a lower probability of exploring the environment. In the limit of zero temperature T m 0, the agent uses the target policy to select greedy actions in order to maximize the obtained rewards."

(Xiao et al., 2025)

Research on AGV Path Planning Based on Improved DQN Algorithm

"When τ is high, the selection probability of all actions tends to be uniform; that is, the strategy is more inclined to random exploration. When τ is lower, the strategy is more inclined to choose the action with higher value; that is, it is closer to a greedy strategy."

(Zhang, 2026)

A Survey of Multi-Armed Bandit Algorithms: From Theoretical Foundations to Modern Applications

"Temperature parameter, > 0 , controls the level of exploration. High temperature makes selection probabilities more uniform -this encourages exploration, while low temperature puts the majority of the probability mass on the arm with the highest estimated mean, this encourages exploitation. As with greedy, a decaying schedule must be used to achieve good asymptotic performance (Vermorel et al., 2005)."

(That et al., 2026)

Maximizing SAGIN coverage with joint HAP/UAV placement, power, bandwidth, and beamforming via deep actor-critic with a temperature function-based policy

"From a theoretical perspective, annealing introduces a temperature schedule that encourages broader exploration at early stages and gradually shifts toward exploitation as the temperature decreases, which helps escape local optima in non-convex landscapes."

(Ameen et al., 2026)

Structured Neuron Pruning in Deep Neural Networks Using Multi-Armed Bandits

"Softmax, also known as Boltzmann exploration, converts empirical rewards into a probability distribution over the candidate neurons [11]. A neuron with a larger empirical reward is assigned a larger probability of being selected, while lower-scoring neurons can still be explored. The selection probability is"

"where v t >0 is the temperature. A larger value of v t gives a flatter distribution and encourages exploration. A smaller value concentrates probability on neurons with high empirical rewards. A decaying temperature schedule is"

"where v 0 and v T are the initial and final temperatures."

Temperature annealing schedule requirements in Boltzmann exploration

The temperature should usually start high enough to allow broad exploration and then be reduced gradually so the policy becomes greedier over time. The key requirement is to anneal slowly enough that every action keeps getting sampled often enough, but not so slowly that learning stays random for too long. (12 sources)

A standard requirement for Boltzmann exploration is that the temperature should decrease over time rather than stay fixed, so the agent moves from broad exploration toward near-greedy action choice as learning improves. High temperature makes actions close to equiprobable, while low temperature puts much more mass on the current best action. Several sources explicitly connect this to simulated annealing and recommend a suitably slow decay, with logarithmic schedules often given as an example of the kind of rate used to preserve convergence properties. (Csaji et al., 2008) (Elfwing et al., 2017) (Cesa-Bianchi et al., 2017) (Singh et al., 2026)

The practical reason for this requirement is simple: if temperature falls too fast, the policy can become nearly greedy before value estimates are reliable, so some actions may stop being sampled enough to correct early mistakes. This is exactly the risk noted in work on Boltzmann exploration: some schedules can make the agent commit too early to a suboptimal action, while convergence results rely on continuing exploration and, in some formulations, on ensuring each action keeps a nontrivial selection probability over time. (Elfwing et al., 2017) (Cesa-Bianchi et al., 2017) (D'Amico-Wong et al., 2024) (Cesa-Bianchi et al., 2017)

At the same time, annealing cannot be arbitrarily slow. If temperature stays high for too long, behavior remains too close to random, convergence can become very slow, and the learning signal from good actions can be diluted by excess exploration. This tradeoff is described both in tabular RL discussions and in more practical papers that note large initial temperatures or no decay can hurt performance by overwhelming exploitation. (Elfwing et al., 2017) (Ma, 2022) (Han et al., 2019)

There is no single universally best schedule. Older and more practical work often uses monotone decay such as logarithmic, exponential, or linear schedules, with parameters chosen so cooling is gentle enough to maintain exploration. Examples include logarithmic decay recommendations, exponential cooling with a factor close to 1, and linear increases in inverse temperature. (Csaji et al., 2008) (Han et al., 2019) (Zhou et al., 2023) (Peri et al., 2026)

But a notable theoretical result is that in stochastic bandits, Boltzmann exploration with any monotone learning-rate or inverse-temperature sequence can still be suboptimal, which is one reason schedule tuning is considered hard in practice. More refined schemes therefore adapt temperature per arm or use non-monotone schedules, and some recent work even uses cyclical or performance-triggered resets rather than pure one-way cooling. (Cesa-Bianchi et al., 2017) (Biswas et al., 2019) (Ara'ujo et al., 2020) (Kuleshov et al., 2014)

So, in synthesis, the requirement is not just “lower temperature over time,” but “lower it at a rate that still preserves enough visitation while letting the policy become greedy in the limit.” That is the core balance

behind convergence-oriented advice for Boltzmann exploration. (Cesa-Bianchi et al., 2017) (Csaji et al., 2008) (Elfwing et al., 2017)

Evidence

(Csaji et al., 2008)

Adaptive Stochastic Resource Control: A Machine Learning Approach

"In order to ensure the convergence of the FQL algorithm, one must guarantee that each cost-to-go estimation be continuously updated. A technique used often to balance between exploration and exploitation is the Boltzmann formula (also called softmin action selection):"

"where τ ≥ 0 is the Boltzmann (or Gibbs) temperature, i is the episode number. It is easy to see that high temperatures cause the actions to be (nearly) equiprobable, low ones cause a greater difference in selection probability for actions that differ in their value estimations"

".It is advised to extend this approach by a variant of simulated annealing (Kirkpatrick et al., 1983) or Metropolis algorithm (Borkar, 1953), which means that τ should be decreased over time, at a suitable, e.g., logarithmic, rate (Singh et al., 2000)."

(Elfwing et al., 2017)

Sigmoid-Weighted Linear Units for Neural Network Function Approximation in Reinforcement Learning

"In order for Boltzmann exploration to achieve convergence to the optimal policy, the temperature should be decreased sufficiently slowly (Singh et al., 2000). If the temperature is lowered too quickly, then some actions might not get sampled often enough. On the other hand, if the temperature is lowered too slowly, then convergence can take a very long time. Furthermore, with a softmax policy there are no absolute differences in action values, only relative differences. This means that the rewards need to be scaled in a way that yields suitable differences in the action values. As a result, both the reward scale and the temperature schedule need to be tuned for good performance. Our version of Q-learning with neural network function approximation uses standard Q-learning updates and assumes convergence for the tabular case if the learning rates are sufficiently small and satisfy the usual stochastic approximation conditions and if all state-action pairs continue to be visited."

(Cesa-Bianchi et al., 2017)

Boltzmann Exploration Done Right

"As noted on several occasions in the literature, finding the right schedule for η t can be very difficult in practice (Kaelbling et al., 1996)(Vermorel et al., 2005). Below, we quantify this difficulty by showing that natural learning-rate schedules may fail to achieve near-optimal regret guarantees. More precisely, they may draw suboptimal arms too much even after having estimated all the means correctly, or commit too early to a suboptimal arm and never recover afterwards"

".we mention that Singh et al. (2000) propose a learning-rate schedule η t for Boltzmann exploration that simultaneously guarantees that all arms will be drawn infinitely often as T goes to infinity, and that the policy becomes greedy in the limit. This property is proven by choosing a learning-rate schedule adaptively to ensure that in each round t, each arm gets drawn with probability at least 1 t , making it similar in spirit to ε-greedy exploration"

".More importantly, as observed by Cesa-Bianchi et al. (2017), the amount of exploration induced by this strategy can be heavily influenced by the range of the rewards, whereas scaling all rewards by a positive constant should not change the intrinsic difficulty of the problem. For example, if all rewards are in [0, 1], then choosing η t = log t makes the strategy nearly equivalent to uniform random sampling, while the same strategy can become nearly greedy if the rewards are in [0] 10]"

".standard convergence results for Q-learning (Jaakkola et al., 1994;Melo, 2001) only require all state-action pairs to be visited infinitely often, and that the learning rates satisfy the Robbins-Monro conditions"

(Singh et al., 2026)

Regret and Sample Complexity of Online Q-Learning via Concentration of Stochastic Approximation with Time-Inhomogeneous Markov Chains

"The parameter b controls the exploration-exploitation tradeoff, with larger values leading to faster decay of the temperature and hence reduced exploration"

". This choice is tuned to optimize the regret bound in our analysis, and logarithmic temperature schedules of this form are known to be optimal in the stochastic multi-armed bandit setting; see, for example, (Cesa-Bianchi et al., 2017)"

". We first analyze Boltzmann Q-learning with decaying temperature and show that its regret depends critically on the suboptimality gap of the MDP: for sufficiently large gaps, the regret is sublinear, while for small gaps it deteriorates and can approach linear

growth"

". Our results show how the exploration parameters, specifically the temperature schedule for the softmax exploration and the sequence n for greedy policies, affect the convergence rate of the learned Q-values to Q "

". We also obtain sample complexity guarantees, with both regret and sample complexity bounds holding with high probability"

". Moreover, our concentration bounds enable us to establish almost sure convergence of the Q-learning iterates, which is nontrivial for online Q-learning with decaying exploration."

(D'Amico-Wong et al., 2024)

Easy as ABCs: Unifying Boltzmann Q-Learning and Counterfactual Regret Minimization

"While BQL converges to the optimal policy in MDPs and in perfect information games (given a suitable temperature schedule) (Cesa-Bianchi et al., 2017)(Singh et al., 2000), it fails to converge to a Nash equilibrium in imperfect-information multi-agent games such as Poker (Brown et al., 2020)."

(Ma, 2022)

Entropy Augmented Reinforcement Learning

"in order to recover to the original MDP, we need to anneal the temperature coefficient. But we also seek to use the augmentation to boost the exploration of the agent, which means we can not anneal it too fast"

". we propose to exponentially drop the temperature every update to control the behavior of the temperature while maintain a sufficient amount of exploration"

". The results of the Figure 3 reveal that:"

"• With decay is more preferred than no decay"

".The reason for that large initial temperature would harm the performance is that the learning signal is overwhelmed by the augmenting term, which causes the agent to be too exploratory to exploit the beneficial experience. And smaller one makes the reward clearer, that could be more easily taken advantage of by the agent. This is known as the tradeoff between exploration and exploitation."

(Han et al., 2019)

A Reinforcement Learning Method for a Hybrid Flow-Shop Scheduling Problem

"The temperature coefficient T is high and Q-value is relatively small at the beginning of the learning, resulting in all the action to be chosen with nearly equal probabilities, which is beneficial for the agent to explore the actions with non-optimal Q-value. As the learning progresses, the temperature coefficient T decreases gradually, and the probability changes with Q(s, a), the probability of adopting random actions decrease, which is instrumental in selecting the optimal action with the largest Q-value. In the later stage of learning, the temperature parameter T tends to 0, Q(s, a) is the largest, and the corresponding action is selected with the largest probability, and the action with the largest Q-value is selected each time, which means the policy change to greedy policy"

".where λ((0, 1)) is the cooling parameter, often set as a decimal close to 1. e is the times of iteration and e 0 is a positive constant. T 0 is the initial value of temperature coefficient T; if set too small, it will cause the local minimum value of the algorithm; on the contrary, the calculation of the algorithm will be increased with a large T 0 ."

(Zhou et al., 2023)

Learning to Scale Temperature in Masked Self-Attention for Image Inpainting

"Manually designed temperature tuning approaches tune the temperature parameter iteratively with pre-designed strategies. Kirkpatrick et al. (Kirkpatrick et al., 1983) introduced a temperature in Simulated Annealing to balance the exploration and exploitation tradeoff in combinatorial optimization, and used exponentially decreasing annealing strategy to tune the temperature. Similar temperature tuning strategy was also used in Boltzmann Machine learning algorithm (Ackley et al., 1985), and Boltzmann exploration [38] in reinforcement learning."

(Peri et al., 2026)

Smart walkers in discrete space

"The parameter α [0, 1], known as the learning rate, determines the extent to which new information overrides prior knowledge. It governs how aggressively the Q-values are updated based on new experiences"

". Finally, the parameter γ [0, 1], known as the discount factor, controls the agent's preference for future rewards over immediate ones"

". We define the agent's policy π(a|s) as a probability distribution over available actions given a state. This latter distribution is obtained by means of the so called Boltzmann exploration paradigm [31], which applies a weighted softmax function S β to the Q-values of the given state"

". where β > 0 is the inverse temperature parameter controlling the trade-off between exploration and exploitation. Larger values of β place more weight on actions with higher Q-values, promoting exploitation, while smaller values yield a more uniform distribution, encouraging exploration"

". As it is customary during training, we apply a linear temperature annealing schedule, where the inverse temperature is progressively increased as the training progresses. Formally, at episode e the inverse temperature is given by"

(Biswas et al., 2019)

Seeker: Real-Time Interactive Search

"To balance exploration and exploitation, one resorts to annealing [1], with an appropriately tuned sequence of learning rate parameters (aka inverse temperature) η k ≥ 0 for each timestep t k"

"Finding the right schedule for η k can be very difficult in practice (Vermorel et al., 2005). In (Cesa-Bianchi et al., 2017), the authors provide an annealing schedule for η k in a standard stochastic multi-armed bandit setting, guaranteeing sublinear regret. Let n j be the number of times arm j has been played up to timestep t k −1 . For some constant C > 0, they set η = n j /C 2 , and sample according to:"

"Equation 4.13 decouples the learning rates of the individual items, and factors-in the uncertainty of the reward estimates. We now have a proper way to sample from a Boltzmann, with convergence guarantees."

(Ara'ujo et al., 2020)

Single-partition adaptive Q-learning

"Boltzmann exploration has its drawbacks, namely the existence of multiple fixed points (Littman, 1996). To deal with this issue, alternative softmax-based operators have been proposed by (Asadi et al., 2016) and Pan et al. (2019). The approach followed in this paper uses the classical softmax function, while keeping track of the best policy found so far, therefore clipping the instability associated with Boltzmann exploration"

".The main challenge involved with the use of Boltzmann exploration is setting the temperature schedule. A straightforward approach, used by (Tijsma et al., 2016), is to set a constant temperature, which can be found using grid search or any other tuning method. Another approach is to decrease (anneal) the temperature as some function of the training iteration (Pan et al., 2019). The approach herein followed is the opposite one: start with a low temperature, and gradually increase it if the performance is not improving. This allows a gradual shift from exploitation to exploration. Once the policy improves, the temperature is reset to a low value, and the procedure is repeated. This is similar to cyclical annealing schedules, which have already been used in supervised learning applications (Smith, 2015)Smith and Topin, 2018;(Huang et al., 2017)(Fu et al., 2019)Mu et al., 2018a;(Loshchilov et al., 2016). In all these methods, the cycle length is controlled by at least one user-defined parameter. In the algorithm proposed in this paper, the cycle reset occurs automatically, when an increase in performance is detected. The user only has to provide two parameters: one parameter controlling the rate at which the temperature heads towards the maximum value; another that works similarly to a doubling period factor."

(Kuleshov et al., 2014)

Algorithms for multi-armed bandit problems

"As in the case of -greedy, polylogarithmic regret bounds exist Cesa-Bianchi and Fischer (1998) for decreasing τ schedules. However, because empirical evidence suggests that such schedules offer no practical advantage (Vermorel et al., 2005), we will use fixed values of τ in our experiments."

What happens if temperature decreases too quickly or too slowly

If temperature drops too fast, the policy can lock onto early noisy value estimates and stop exploring enough to recover. If it drops too slowly, the agent keeps taking weak actions for too long, so learning and reward collection both become inefficient. (7 sources)

The main failure mode when temperature decreases too quickly is premature commitment. Early estimation noise can make one action look best by accident, and a fast drop in temperature then sharply reduces the chance of trying alternatives, so the agent can get stuck in a suboptimal policy or local optimum before it has enough evidence to correct itself. This is a standard warning in both RL and annealing-based discussions, and it is also why overly aggressive schedules can fail the conditions needed for proper long-run exploration.

(Elfwing et al., 2013) (Reverdy et al., 2013) (Martinez-Rios et al., 2012) (O'Donoghue, 2018) (Overman et al., 2026) (Cesa-Bianchi et al., 2017)

If temperature decreases too slowly, the problem is the opposite: the policy stays too stochastic. The agent keeps sampling actions that are already known to be worse, so convergence slows down and finite-time regret or exploration cost rises. In simple terms, the learner keeps “wasting” decisions on exploration after the useful part of exploration should have ended. (Elfwing et al., 2013) (Reverdy et al., 2013) (O'Donoghue, 2018)

A useful nuance is that this tradeoff becomes especially hard when action values are close together. In that case, even modest temperature changes can swing the policy a lot, and tuning becomes delicate: cool too fast and small noisy gaps get over-trusted; cool too slowly and the learner keeps dithering among near-tied actions. Kaelbling et al. note that softmax works better when the best action is clearly separated and otherwise may require careful manual tuning to avoid very slow convergence. (Kaelbling et al., 1996)

So the practical picture is: fast cooling risks under-exploration and bad lock-in, while slow cooling risks over-exploration and slow improvement. The “right” schedule is the one that keeps enough exploration long enough to fix early mistakes, but still lets exploration fade soon enough for the policy to concentrate on the best actions within a useful time horizon. (Elfwing et al., 2013) (Reverdy et al., 2013)

Evidence

(Elfwing et al., 2013)

Scaled free-energy based reinforcement learning for robust and efficient learning in high-dimensional state spaces

> *"For successful convergence of Boltzmann exploration, the temperature τ has to be lowered slowly enough for all state-action pairs to be visited infinitely often, and at the same time quickly enough for exploration to stop eventually and to avoid convergence to suboptimal asymptotic behavior (Singh et al., 2000). If the temperature decreases too quickly, the system may get stuck in a suboptimal policy, and if it decreases too slowly, the system may converge very slowly and incur unnecessary exploration costs"*
>
> *". It should be noted that softmax action selection is not invariant to reward scaling, because a large reward scale relative to the temperature makes the policy nearly deterministic, whereas a small reward scale makes it nearly uniform"*
>
> *". For convergence of Q-learning, the learning rates must satisfy the usual stochastic approximation conditions, and all state-action pairs must continue to be visited infinitely often."*

(Reverdy et al., 2013)

Modeling Human Decision Making in Generalized Gaussian Multiarmed Bandits

> *"good cooling schedules for simulated annealing take the form υ t = v/ log t, where v > 0 is chosen sufficiently large for convergence. If the temperature decreases too quickly, then the probability of selecting an arm other than the current argmax decreases too quickly, and one gets trapped in local optima. On the other hand, if the temperature decreases too slowly, the algorithm is too stochastic and converges slowly.*
>
> *The quality of a softmax policy depends strongly on the scale of the rewards, because multiplying all rewards by a constant is equivalent to dividing the temperature by that constant.*
>
> *For Q-learning, convergence assumptions typically require that the learning rates satisfy the Robbins-Monro conditions and that all state-action pairs are visited infinitely often."*

(Martinez-Rios et al., 2012)

A Simulated Annealing Algorithm for the Satisfiability Problem Using Dynamic Markov Chains with Linear Regression Equilibrium

> *"Some criticisms about Simulated Annealing are about the long time of execution of standard Boltzmann-type Simulated Annealing, has many times driven these projects to utilize a temperature schedule too fast to satisfy the sufficiency conditions required to establish a true ergodic search."*

(O'Donoghue, 2018)

Variational Bayesian Reinforcement Learning with Regret Bounds

"Boltzmann exploration can produce suboptimal behavior if the temperature annealing schedule is not chosen carefully. If the temperature decreases too quickly, then some state-action pairs may not be explored enough and the algorithm may fail to converge to the optimal policy. If the temperature decreases too slowly, then the policy may continue to execute bad actions. This is undesirable because regret, unlike PAC sample complexity, concerns finite elapsed time T . For this reason, much of the practical use of Boltzmann exploration is in settings where there is no commitment to eventual asymptotic convergence.

Perhaps the main issue with Boltzmann exploration is that the policy can be very sensitive to the scale of the rewards. Consider a stochastic multi-armed bandit problem where the rewards are very small compared to the horizon. In this case, the softmax policy is nearly uniform and it can take an extremely long time to identify the best arm, no matter what schedule is chosen.

Q-learning is perhaps the most popular RL algorithm due to its simplicity [39]. Convergence of Q-learning was established in [30] assuming every state-action pair is visited infinitely often and the learning rates satisfy standard stochastic approximation assumptions."

(Overman et al., 2026)

Annealed Softmax Greedy in Many-Armed Bayesian Bandits

"A classical result of (Cesa-Bianchi et al., 2017) shows that this type of exploration (also referred to as Boltzmann exploration) can suffer linear regret with a small number of arms under any monotone temperature schedule, and that strong guarantees in that regime typically require schedules that adapt to arm-specific information. The standard failure mode is "cooling too fast": early noise gets amplified, causing premature concentration on suboptimal actions."

(Cesa-Bianchi et al., 2017)

Boltzmann Exploration Done Right

"As noted on several occasions in the literature, finding the right schedule for η t can be very difficult in practice (Kaelbling et al., 1996)(Vermorel et al., 2005). Below, we quantify this difficulty by showing that natural learning-rate schedules may fail to achieve near-optimal regret guarantees. More precisely, they may draw suboptimal arms too much even after having estimated all the means correctly, or commit too early to a suboptimal arm and never recover afterwards"

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".More importantly, as observed by Cesa-Bianchi et al. (2017), the amount of exploration induced by this strategy can be heavily influenced by the range of the rewards, whereas scaling all rewards by a positive constant should not change the intrinsic difficulty of the problem. For example, if all rewards are in [0, 1], then choosing η t = log t makes the strategy nearly equivalent to uniform random sampling, while the same strategy can become nearly greedy if the rewards are in [0] 10]"

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(Kaelbling et al., 1996)

Reinforcement Learning: A Survey

"The temperature parameter T can be decreased over time to decrease exploration. This method works well if the best action is well separated from the others, but su ers somewhat when the values of the actions are close. It may also converge unnecessarily slowly unless the temperature schedule is manually tuned with great care."

How reward scaling affects softmax policies

Softmax policies depend on the size of the Q-value differences relative to temperature, so changing reward scale can strongly change behavior even if the task itself is the same. Larger reward scales make the policy more greedy at a fixed temperature, while smaller scales make it more random. (6 sources)

A key practical issue with Boltzmann or softmax exploration is that it is **not invariant to reward scaling**. If all rewards, and therefore learned Q-values, are multiplied by a constant, the softmax probabilities change as if the temperature had been divided by that same constant. So larger reward magnitudes relative to temperature make the policy much sharper and closer to greedy, while smaller reward magnitudes make it flatter and closer to uniform random choice. This is stated directly in several sources and is one reason softmax often needs both reward normalization and temperature tuning in practice. (Elfwing et al., 2013)

(Reverdy et al., 2013) (Elfwing et al., 2017)

This matters because softmax uses **relative differences** in action values, not absolute notions of “good” or “bad.” If value gaps are tiny on the scale set by temperature, the policy will barely prefer the better action. If the gaps are large, even a moderate temperature can make the policy almost deterministic. In that sense, reward scaling effectively changes how much exploration the same nominal temperature produces. (Elfwing et al., 2017) (Elfwing et al., 2013)

The consequence is that the same annealing rule can behave very differently across problems with different reward ranges. A notable example from the literature is that a schedule such as \eta_t=\log t can act almost like uniform sampling when rewards lie in , yet become nearly greedy when rewards lie in a much larger range, even though the intrinsic decision problem has not changed. This sensitivity is one reason schedule design for Boltzmann exploration is considered difficult in practice. (Cesa-Bianchi et al., 2017) (Kaelbling et al., 1996)

In bandit settings, this can be especially harmful when rewards are very small relative to the horizon or to the chosen temperature scale: the softmax policy can stay close to uniform for a very long time and take too long to identify the best action. More generally, if rewards are scaled too small, exploration is excessive; if they are scaled too large, the policy can become prematurely greedy and lose the benefit of soft exploration. (O'Donoghue, 2018) (Elfwing et al., 2013) (Reverdy et al., 2013)

So the main takeaway is that with softmax, **temperature has no meaning by itself**; it only has meaning relative to the scale of rewards or learned Q-values. In practice, that means one often needs to tune temperature together with reward scaling, reward clipping, or value normalization if one wants similar exploration behavior across tasks. (Elfwing et al., 2017) (Cesa-Bianchi et al., 2017)

Evidence

(Elfwing et al., 2013)

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> *". For convergence of Q-learning, the learning rates must satisfy the usual stochastic approximation conditions, and all state-action pairs must continue to be visited infinitely often."*

(Reverdy et al., 2013)

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> *"good cooling schedules for simulated annealing take the form υ t = v/ log t, where v > 0 is chosen sufficiently large for convergence. If the temperature decreases too quickly, then the probability of selecting an arm other than the current argmax decreases too quickly, and one gets trapped in local optima. On the other hand, if the temperature decreases too slowly, the algorithm is too stochastic and converges slowly.*
>
> *The quality of a softmax policy depends strongly on the scale of the rewards, because multiplying all rewards by a constant is equivalent to dividing the temperature by that constant.*
>
> *For Q-learning, convergence assumptions typically require that the learning rates satisfy the Robbins-Monro conditions and that all state-action pairs are visited infinitely often."*

(Elfwing et al., 2017)

Sigmoid-Weighted Linear Units for Neural Network Function Approximation in Reinforcement Learning

"In order for Boltzmann exploration to achieve convergence to the optimal policy, the temperature should be decreased sufficiently slowly (Singh et al., 2000). If the temperature is lowered too quickly, then some actions might not get sampled often enough. On the other hand, if the temperature is lowered too slowly, then convergence can take a very long time. Furthermore, with a softmax policy there are no absolute differences in action values, only relative differences. This means that the rewards need to be scaled in a way that yields suitable differences in the action values. As a result, both the reward scale and the temperature schedule need to be tuned for good performance. Our version of Q-learning with neural network function approximation uses standard Q-learning updates and assumes convergence for the tabular case if the learning rates are sufficiently small and satisfy the usual stochastic approximation conditions and if all state-action pairs continue to be visited."

(Cesa-Bianchi et al., 2017)

Boltzmann Exploration Done Right

"As noted on several occasions in the literature, finding the right schedule for η t can be very difficult in practice (Kaelbling et al., 1996)(Vermorel et al., 2005). Below, we quantify this difficulty by showing that natural learning-rate schedules may fail to achieve near-optimal regret guarantees. More precisely, they may draw suboptimal arms too much even after having estimated all the means correctly, or commit too early to a suboptimal arm and never recover afterwards"

".we mention that Singh et al. (2000) propose a learning-rate schedule η t for Boltzmann exploration that simultaneously guarantees that all arms will be drawn infinitely often as T goes to infinity, and that the policy becomes greedy in the limit. This property is proven by choosing a learning-rate schedule adaptively to ensure that in each round t, each arm gets drawn with probability at least 1 t , making it similar in spirit to ε-greedy exploration"

".More importantly, as observed by Cesa-Bianchi et al. (2017), the amount of exploration induced by this strategy can be heavily influenced by the range of the rewards, whereas scaling all rewards by a positive constant should not change the intrinsic difficulty of the problem. For example, if all rewards are in [0, 1], then choosing η t = log t makes the strategy nearly equivalent to uniform random sampling, while the same strategy can become nearly greedy if the rewards are in [0] 10]"

".standard convergence results for Q-learning (Jaakkola et al., 1994;Melo, 2001) only require all state-action pairs to be visited infinitely often, and that the learning rates satisfy the Robbins-Monro conditions"

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Reinforcement Learning: A Survey

"The temperature parameter T can be decreased over time to decrease exploration. This method works well if the best action is well separated from the others, but su ers somewhat when the values of the actions are close. It may also converge unnecessarily slowly unless the temperature schedule is manually tuned with great care."

(O'Donoghue, 2018)

Variational Bayesian Reinforcement Learning with Regret Bounds

"Boltzmann exploration can produce suboptimal behavior if the temperature annealing schedule is not chosen carefully. If the temperature decreases too quickly, then some state-action pairs may not be explored enough and the algorithm may fail to converge to the optimal policy. If the temperature decreases too slowly, then the policy may continue to execute bad actions. This is undesirable because regret, unlike PAC sample complexity, concerns finite elapsed time T . For this reason, much of the practical use of Boltzmann exploration is in settings where there is no commitment to eventual asymptotic convergence.

Perhaps the main issue with Boltzmann exploration is that the policy can be very sensitive to the scale of the rewards. Consider a stochastic multi-armed bandit problem where the rewards are very small compared to the horizon. In this case, the softmax policy is nearly uniform and it can take an extremely long time to identify the best arm, no matter what schedule is chosen.

Q-learning is perhaps the most popular RL algorithm due to its simplicity [39]. Convergence of Q-learning was established in [30] assuming every state-action pair is visited infinitely often and the learning rates satisfy standard stochastic approximation assumptions."

Q-learning convergence assumptions: learning rates and visitation

Standard Q-learning convergence results assume two core things: every state-action pair keeps getting visited infinitely often, and the learning rates satisfy the usual Robbins-Monro stochastic approximation conditions. In this context, Boltzmann exploration is only acceptable if its temperature schedule preserves enough exploration to meet that visitation requirement while still becoming greedy in the limit. (4 sources)

For the convergence of tabular Q-learning, the notable theoretical assumptions are quite simple to state. First, **all state-action pairs must be visited infinitely often**. Second, the **learning rates must satisfy the**

standard stochastic approximation or Robbins-Monro conditions rather than staying too large or vanishing too fast. The provided sources summarize standard results this way and explicitly point back to classic convergence analyses. (Elfwing et al., 2013) (Cesa-Bianchi et al., 2017) (O'Donoghue, 2018) (Elfwing et al., 2017)

In practical terms, the Robbins-Monro conditions mean the step sizes should usually **decrease over time**, with enough total learning to keep correcting estimates but not so much persistent noise that the values fail to settle. A common textbook form is that, for each repeatedly updated state-action pair, the learning rates \alpha_t satisfy \sum_t \alpha_t = \infty and \sum_t \alpha_t^2 < \infty. (Model-Generated)

This is why the exploration policy matters for Q-learning convergence. If Boltzmann temperature falls too fast, some actions may stop being tried often enough, breaking the infinite-visitation assumption and allowing early value errors to persist. If it falls slowly enough that each action keeps a lower-bounded probability of being sampled, then it can support the visitation condition while the policy gradually becomes greedy. (Elfwing et al., 2013) (Cesa-Bianchi et al., 2017) (O'Donoghue, 2018) (Elfwing et al., 2017)

A useful way to connect this back to the earlier discussion is: **Q-learning convergence theory does not require Boltzmann exploration specifically**. What it requires is enough continued exploration to ensure infinite visitation, plus valid learning-rate decay. So Boltzmann exploration is compatible with Q-learning convergence only when its annealing schedule is chosen to preserve those assumptions; otherwise, the algorithm can lose the standard guarantee even if the update rule itself is unchanged. (Cesa-Bianchi et al., 2017) (O'Donoghue, 2018) (Elfwing et al., 2017)

Evidence

(Elfwing et al., 2013)

Scaled free-energy based reinforcement learning for robust and efficient learning in high-dimensional state spaces

"For successful convergence of Boltzmann exploration, the temperature τ has to be lowered slowly enough for all state-action pairs to be visited infinitely often, and at the same time quickly enough for exploration to stop eventually and to avoid convergence to suboptimal asymptotic behavior (Singh et al., 2000). If the temperature decreases too quickly, the system may get stuck in a suboptimal policy, and if it decreases too slowly, the system may converge very slowly and incur unnecessary exploration costs"

". It should be noted that softmax action selection is not invariant to reward scaling, because a large reward scale relative to the temperature makes the policy nearly deterministic, whereas a small reward scale makes it nearly uniform"

". For convergence of Q-learning, the learning rates must satisfy the usual stochastic approximation conditions, and all state-action pairs must continue to be visited infinitely often."

(Cesa-Bianchi et al., 2017)

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